

Modeling the disaster relief hub network design problem: A case of Central Vietnam

Phu-Quy Le-Tang¹, Quang-Vu Nguyen¹, and Sunarin Chanta^{2*}

¹ The University of Danang, Vietnam - Korea University of Information and Communication Technology

² King Mongkut's University of Technology North Bangkok
quyltp.22git@vku.udn.vn, nquvu@vku.udn.vn, sunarin.c@itm.kmutnb.ac.th

Abstract. Natural disasters severely disrupt transportation infrastructure, heavily complicating search and rescue (SAR) operations. Existing humanitarian logistics models often fail to capture these dynamic disruptions and the non-linear escalation of human suffering. To address this gap, we propose a two-stage stochastic Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP). Our model simultaneously minimizes expected logistics costs – utilizing Daganzo’s Continuous Approximation for scalable last-mile routing – and the expected maximum deprivation cost to enforce social equity. To solve this NP-hard problem, we develop a Priority-Based Non-dominated Sorting Genetic Algorithm II (PB-NSGA-II). Extensive computational experiments on standard benchmarks (AP, TR81) validate the algorithm’s efficiency. Furthermore, a detailed case study of the flood-prone Vu Gia – Thu Bon river basin provides crucial managerial insights. The results highlight the necessity of dynamically adapting multi-modal network topologies to maintain resilience under extreme disaster scenarios.

Keywords: Humanitarian Logistics · Disaster relief · Hub location problem · Multi-objective Evolutionary Algorithm.

1 Introduction

1.1 Problem

Natural disasters are becoming increasingly frequent and violent, continuously exposing supply chain weaknesses and causing massive economic losses [2, 4]. As a highly vulnerable coastal country, Vietnam has witnessed unprecedented extreme weather events. In 2025 alone, a relentless sequence of typhoons and record-breaking floods in Central Vietnam resulted in massive fatalities and an estimated 98.7 trillion VND in economic losses [12, 15].

There were many great challenges in humanitarian operations during this period, one of them being the severe disruption of national logistics infrastructure. The North-South Railway – backbone of Vietnam’s transport network was

* Corresponding author.

paralyzed due to track erosion and submersion, forcing the cancellation of 170 trains [12]. Simultaneously, key arteries such as the National Highway 1 and routes in the Central Highlands were severed by landslides, isolating communities for days [10]. These events demonstrate that during a major catastrophe, the assumption of a fully connected transport network is fundamentally flawed. The physical network becomes “incomplete”, characterized by broken links and isolated nodes. Another challenge is resource scarcity resulting from poor planning and uncertain supply and demand. Disaster management involves four phases: mitigation, preparedness, response, and recovery [1]. Disaster Relief Network Design (DRND) bridges the preparedness and response phases by optimizing facility locations and relief routing.

1.2 Related Works

The design of disaster relief networks differs fundamentally from commercial logistics. Commercial models prioritize profit maximization. Humanitarian logistics prioritizes human lives and rapid response [14]. Facility location is a critical component of disaster preparedness. A comprehensive review by Boonmee et al. [3] highlights the growing need for facility location models under severe uncertainty.

Disruption and Network Design. Disasters often cause severe infrastructure disruptions. Standard network models fail when paths become unavailable. Yahyaei and Bozorgi-Amiri [16] addressed this by proposing a robust reliable network design. They integrated shelter and supply facility locations under disruption risks. Similarly, Hasani and Mokhtari [8] proposed an integrated relief network model for Iran. They modeled dynamic uncertainty and employed multi-objective optimization. However, embedding exact routing sequences into these network designs creates highly intractable Location-Routing Problems (LRP).

Continuous Approximation. To circumvent the NP-hard complexity of exact routing, researchers utilize approximations. Daganzo [5] pioneered the Continuous Approximation (CA) method. CA estimates last-mile routing costs using spatial densities rather than explicit node-to-node sequences. This technique drastically reduces computational burden. It allows large-scale network models to remain tractable. Our model adopts this CA framework to estimate evacuation costs efficiently.

Deprivation Cost and Social Equity. Traditional objective functions usually minimize pure logistics costs or average travel times. Holguín-Veras et al. [9] argued that linear time penalties neglect social equity. They introduced the concept of deprivation cost. It applies an exponential penalty function to waiting times. This mathematically captures the non-linear escalation of human suffering during disasters. Incorporating deprivation cost forces optimization models to distribute resources equitably. It prevents the marginalization of isolated distress clusters.

Solution Approaches. Disaster relief models often involve multiple conflicting objectives. Meta-heuristics are highly preferred for these problems. NSGA-II [6] is a standard and robust evolutionary framework. It utilizes non-dominated

sorting and crowding distance to find Pareto-optimal solutions. Recent studies frequently adapt NSGA-II for humanitarian operations.

Research Gap and Our Contribution. Existing literature contains a significant gap. Very few studies integrate incomplete hub networks, Daganzo’s continuous approximation, and exponential deprivation costs under a two-stage stochastic framework. Solving such a combined model is computationally prohibitive for exact solvers. Our study addresses this gap. We formulate the MO-IHLNDP and propose a meta-heuristic algorithm named Priority-Based NSGA-II, which can handle large-scale instances effectively.

2 Problem Description and Mathematical Model

To design an efficient disaster relief network (DRND), specifically adjusted for flood disaster response in Central Vietnam, we framed the problem as a multi-objective incomplete hub location network design problem (MO-IHLNDP). This model is the capacitated, single-allocation variant of the HLP.

The main components of our network are origins, hubs, and demands. In our practical settings, hubs refer to the rescue stations. There are two kinds of hubs, corresponding to two kinds of strategies. The first type is established in advance of the disasters, strategically positioned, and inventoried with amenities. The second type is opened dynamically for disaster response and has high approachability to affected areas. Lastly, during the disaster response, we gradually discover the victim groups, which are referred to as **demands**. Besides the government’s planned forces, some volunteers spontaneously offer to help; these are referred to as **origins**.

From a practical point of view, DRND involves multiple critical uncertainty factors. Firstly, the disruption of transportation infrastructure (e.g., damaged roads and buildings) can easily disable pre-programmed plans, urging the need for a dynamic solution [16]. Secondly, the exact locations of the demands, the total number of victims, and the availability of supply origins are highly uncertain during the rescue process [13].

To account for these uncertainties, we employ a two-stage stochastic programming approach utilizing multiple scenarios. In each scenario, a specific risk factor is assigned to each area. We also assume that search and rescue (SAR) operations rely heavily on dynamic external supply sources, as pre-positioned hubs may not fully satisfy the uncertain demands under extreme scenarios.

Finally, the model places a strong emphasis on *social equity* by integrating the deprivation cost [9]. To obtain a robust approximation of the rescue time from an established hub to its assigned demand points without causing computational intractability, we utilize Daganzo’s continuous approximation formula [5], a technique widely adopted in routing problems.

2.1 Sets and Indices

$\mathcal{H}, \mathcal{I}, \mathcal{J}$	Set of candidate hubs (k, h), demand nodes (i), and origins (j)
$\mathcal{S}, \mathcal{M}$	Set of disaster scenarios (s) and transportation modes (m)
$\mathcal{V}$	Set of all nodes (u, v) representing the area in question, $\mathcal{V} = \mathcal{H} \cup \mathcal{I} \cup \mathcal{J}$

2.2 Parameters

First-Phase Parameters (Strategic Planning)

F_k	Fixed cost to establish hub k of the first type (before the disaster)
c_k, κ_k	Unit holding cost and capacity limit for inventory at hub k
α, χ, M	Discount factor for economies of scale, max acceptable risk threshold, and Big-M constant
C_{uvm}, τ_{uvm}	Unit transportation cost and estimated travel time from node u to v via mode m
C_m, Q_m	Unit local routing cost and transportation capacity for vehicle mode m
γ	Conversion factor (average amount of relief items required per evacuated person)

Second-Phase Parameters (Disaster Response, Scenario s)

π_s	Probability of scenario s
F_{ks}^a	Fixed cost to establish hub k of the second type (reactive) in scenario s
τ_{ks}	Processing time for each rescue round trip at hub k in scenario s
$a_{uvm s}$	Accessibility of arc (u, v) via mode m in scenario s (1 if traversable, 0 otherwise)
r_{us}, λ_{is}	Risk index of node u , and deprivation sensitivity coefficient for demand i in scenario s
D_{is}, O_{js}	Demand (number of people to evacuate) at i , and relief items provided at origin j in scenario s
Θ_{kis}	Pre-computed CA routing cost to execute the evacuation from grid i to hub k in scenario s
Φ, η, A_i	Daganzo's circuitry factor, average group size per distress location, and area of grid cell i

2.3 Decision Variables

$x_k \in \{0, 1\}$	1 if hub k is established as the first type (planned), 0 otherwise
$y_{ks} \in \{0, 1\}$	1 if hub k is established as the second type (reactive) in scenario s , 0 otherwise
$z_{iks} \in \{0, 1\}$	1 if demand i is assigned to be evacuated to hub k in scenario s , 0 otherwise
$z_{jks} \in \{0, 1\}$	1 if origin j supplies hub k in scenario s , 0 otherwise
$q_k \geq 0$	Amount of relief items inventoried at hub k
$f_{khms} \geq 0$	Lateral trans-shipment flow volume from hub k to hub h via mode m in scenario s

2.4 Objectives

Objective 1: Minimize Total Expected Logistics Cost (Z_1)

$$\begin{aligned}
\text{Min } Z_1 = & \underbrace{\sum_{k \in \mathcal{H}} F_k x_k + \sum_{k \in \mathcal{H}} c_k q_k}_{\text{Pre-disaster strategic costs}} \\
& + \sum_{s \in \mathcal{S}} \pi_s \left[\underbrace{\sum_{k \in \mathcal{H}} F_{ks}^a y_{ks}}_{\text{Reactive setup}} + \underbrace{\sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{H}} \left(\min_{m \in \mathcal{M} | a_{jkm} = 1} C_{jkm} \right) O_{js} z_{jks}}_{\text{Origin to Hub supply flow}} \right. \\
& \left. + \underbrace{\sum_{k \in \mathcal{H}} \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} \alpha C_{khm} f_{khms}}_{\text{Inter-hub lateral trans-shipment}} + \underbrace{\sum_{k \in \mathcal{H}} \sum_{i \in \mathcal{I}} \theta_{kis} z_{iks}}_{\text{Grid-based Last-Mile CA}} \right] \quad (1)
\end{aligned}$$

Objective Z_1 minimizes the total expected logistics cost, balancing pre-disaster investments against reactive setups and transportation costs. To have an accurate estimation of last-mile evacuation cost while keeping the model simple, we adopted Daganzo's Continuous Approximation [5]. The region of interest is partitioned via spatial discretization, and the demands are aggregated on a grid basis. The routing cost θ_{kis} is pre-computed as a parameter:

$$\theta_{kis} = \min_{m \in \mathcal{M}} \left(2 \cdot C_{kim} \cdot \lceil D_{is}/Q_m \rceil + C_m \cdot \Phi \sqrt{\lceil D_{is}/\eta \rceil \cdot A_i} \right)$$

Objective 2: Minimize Expected Maximum Deprivation Cost (Z_2)

$$\text{Min } Z_2 = \sum_{s \in \mathcal{S}} \pi_s \left(\max_{i \in \mathcal{I}} [D_{is} \cdot (e^{\lambda_{is} \cdot \Omega_{is}} - 1)] \right) \quad (2)$$

The second objective ensures social equity by minimizing the expected maximum deprivation cost across all scenarios. Based on Holguín-Veras et al. [9], Z_2 adopts an exponential function to penalize long waiting times (Ω_{is}) for victims. Furthermore, it forces the model to distribute resources evenly, ensuring overall low delay time for everyone.

$$\Omega_{is} = \sum_{k \in \mathcal{H}} z_{iks} \left(\tau_{ks} + 2 \cdot \min_{m \in \mathcal{M}} \tau_{kim} \right)$$

The sensitivity coefficient λ_{is} is proportional to the risk index r_{is} ($\lambda_{is} = \lambda_0(1 + r_{is})$), effectively prioritizing vulnerable populations. Although Z_2 introduces non-linearity, our meta-heuristic solution approach can directly evaluate it as a fitness function. For linearization into a Mixed-Integer Linear Programming (MILP) form, one can exploit the single-allocation property ($\sum_{k \in \mathcal{H}} z_{iks} = 1$).

2.5 Constraints

The complete MO-IHLNDP model is formally stated as follows:

$$\text{Minimize } Z_1, Z_2$$

$$\text{subject to Constraints (3) – (16)}$$

$$x_k + y_{ks} \leq 1 \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (3)$$

$$q_k \leq \kappa_k \cdot x_k \quad \forall k \in \mathcal{H} \quad (4)$$

$$r_{ks} \cdot (x_k + y_{ks}) \leq \chi \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (5)$$

$$\sum_{k \in \mathcal{H}} z_{iks} = 1 \quad \forall i \in \mathcal{I}, s \in \mathcal{S} \quad (6)$$

$$\sum_{k \in \mathcal{H}} z_{jks} = 1 \quad \forall j \in \mathcal{J}, s \in \mathcal{S} \quad (7)$$

$$z_{iks} \leq x_k + y_{ks} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (8)$$

$$z_{jks} \leq x_k + y_{ks} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (9)$$

$$z_{iks} \leq \sum_{m \in \mathcal{M}} a_{ikms} \quad \forall i \in \mathcal{I}, k \in \mathcal{H}, s \in \mathcal{S} \quad (10)$$

$$z_{jks} \leq \sum_{m \in \mathcal{M}} a_{jkms} \quad \forall j \in \mathcal{J}, k \in \mathcal{H}, s \in \mathcal{S} \quad (11)$$

$$f_{khms} \leq M \cdot a_{khms} \quad \forall k, h \in \mathcal{H}, m \in \mathcal{M}, s \in \mathcal{S} \quad (12)$$

$$\begin{aligned} \sum_{i \in \mathcal{I}} \gamma D_{is} z_{iks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{khms} &\leq q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} \\ &+ \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \end{aligned} \quad (13)$$

$$q_k + \sum_{j \in \mathcal{J}} O_{js} z_{jks} + \sum_{h \in \mathcal{H}} \sum_{m \in \mathcal{M}} f_{hkms} \leq \kappa_k \cdot (x_k + y_{ks}) \quad \forall k \in \mathcal{H}, s \in \mathcal{S} \quad (14)$$

$$x_k, y_{ks}, z_{iks}, z_{jks} \in \{0, 1\} \quad \forall i, j, k, s \quad (15)$$

$$q_k, f_{khms} \geq 0 \quad \forall k, h, m, s \quad (16)$$

Constraints (3) define the hub establishment. For planned hubs, inventory is bounded by physical capacity (4), while the reactive hubs are required to be placed within safe zones (5). Constraints (6)-(11) rigorously enforce the single-allocation property to active hubs via available links. Constraint (12) restricts flow to be trans-shipped on intact paths. Constraints (13) and (14) maintain flow conservation and capacity limits. Specifically, (13) ensures that total supply, comprising inventory (q_k), external origins (O_{js}), and incoming flows – satisfies both victim demand (converted by γ) and outgoing trans-shipments. Variable domains are defined in (15)-(16).

3 Proposed Algorithm: PB-NSGA

Complexity motivation. The MO-IHLNDP is NP-hard: its single-scenario, single-objective relaxation subsumes the uncapacitated facility location problem (UFLP), which is NP-complete [?]. The two-stage stochastic structure further couples exponentially many scenario-dependent second-stage variables to shared first-stage decisions; the bi-objective formulation eliminates any single-objective exact-solver strategy. Exact enumeration is feasible only for small instances ($|\mathcal{H}| \leq 10$, Section 4.3); practical-scale instances require a meta-heuristic.

We therefore propose **PB-NSGA** (Priority-Based Nondominated Sorting Genetic Algorithm), embedding the NSGA-II multi-objective evolutionary framework [6] with a custom *priority-based encoder-decoder* scheme. NSGA-II approximates the Pareto front through fast non-dominated sorting and crowding-distance selection without requiring objective weights – well-suited for bi-objective problems with competing cost and deprivation objectives. The key innovation is that the chromosome encodes only first-stage strategic decisions; all scenario-dependent second-stage variables are reconstructed per scenario by a structured heuristic decoder. This reduces chromosome length from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}| + |\mathcal{I}|)$, yielding feasible decodings at every generation.

3.1 Algorithm Overview

Algorithm ?? presents the complete PB-NSGA procedure. A population of N chromosomes is initialised at random; in each generation, N offspring are produced via genetic operators, the combined $2N$ pool is ranked by fast non-dominated sorting, and the best N individuals survive – with Hamming diversity over hub-opening vectors $\mathbf{X}$ as a tiebreaker to preserve genotypic variety.

3.2 Chromosome Encoding Scheme

In GA [?], a chromosome encodes the decision variables (thus, the implied solution) in its representation. In PB-NSGA, only first-stage strategic decisions are carried in the chromosome and optimized via evolution, while second-stage variables are reconstructed for each scenario $s \in \mathcal{S}$ using the heuristic decoder.

Specifically, each chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$ consists of four encoded segments. The binary vector $\mathbf{X} \in \{0, 1\}^{|\mathcal{H}|}$ determines the establishment of planned hubs ($X_k = 1$). The continuous vector $\mathbf{R} \in [0, 1]^{|\mathcal{H}|}$ defines the inventory fill fraction, setting the pre-positioned inventory as $q_k = R_k \cdot \kappa_k$. The integer vector $\mathbf{A} \in \{0, \dots, |\mathcal{H}| - 1\}^{|\mathcal{I}|}$ defines the preferred group of proximate hubs for each demand i . Lastly, the continuous vector $\mathbf{W} \in [0, 1]^6$ contains six heuristic weights governing the decoder’s behavior.

w_0 and w_3 dictate the demand sorting priority, favoring demands with high urgency ($\lambda \cdot D$) and severe isolation (fewer reachable hubs). During the assignment phase, w_1 , w_2 , and w_4 score the candidate hubs by prioritizing faster travel times, higher residual inventory, and a preference for planned hubs over reactive

Algorithm 1 PB-NSGA for MO-IHLNDP

Require: Population size N , generations G , instance data**Ensure:** Approximate Pareto front $\mathcal{P}$

```
1: Initialise  $P_0$ :  $N$  chromosomes  $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$  drawn at random
2: for each  $\mathbf{C} \in P_0$  do
3:   Evaluate  $(Z_1, Z_2, CV)$ : invoke Alg. ?? for each  $s \in \mathcal{S}$ ; aggregate
4: end for
5: for  $g = 1$  to  $G$  do
6:    $Q_g \leftarrow \text{GENERATEOFFSPRING}(P_{g-1})$ : binary tournament, crossover, mutation
7:   for each  $\mathbf{C} \in Q_g$  do
8:     Evaluate  $(Z_1, Z_2, CV)$ 
9:   end for
10:   $R_g \leftarrow P_{g-1} \cup Q_g$   $\{|R_g| = 2N\}$ 
11:   $\mathcal{F}_1, \mathcal{F}_2, \dots \leftarrow \text{FASTNONDOMINATEDSORT}(R_g)$ 
12:   $P_g \leftarrow \emptyset$ ;  $i \leftarrow 1$ 
13:  while  $|P_g| + |\mathcal{F}_i| \leq N$  do
14:     $P_g \leftarrow P_g \cup \mathcal{F}_i$ ;  $i \leftarrow i + 1$ 
15:  end while
16:  Fill remainder from  $\mathcal{F}_i$  by crowding distance (tiebreak: Hamming distance in  $\mathbf{X}$ )
17: end for
18: return  $\mathcal{P} \leftarrow$  non-dominated solutions in  $P_G = 0$ 
```

ones. Finally, w_5 controls the conservatism towards opening new reactive hubs. This approach effectively reduces the search space from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}|)$.

The four segments are summarised in Table 1; heuristic weight semantics are detailed in Table 2.

Table 1. Chromosome segment definitions.

Segment Type		Size Semantics	
X	Binary	$ \mathcal{H} $	$X_k = 1$: open planned hub k before the disaster
R	$[0, 1]$	$ \mathcal{H} $	Inventory fill fraction: $q_k = R_k \cdot \kappa_k$
A	$\{0, \dots, \mathcal{H} - 1\}$	$ \mathcal{I} $	Preferred anchor hub for demand i ; trial order = hubs by proximity to hub A_i
W	$[0, 1]^6$	6	Heuristic weights governing decoder behaviour

3.3 Priority-Based Decoding Heuristic

The decoder is invoked once per chromosome per scenario $s \in \mathcal{S}$, reconstructing all second-stage decisions from $\mathbf{C}$. It proceeds in four steps.

Step 1 – Initialisation. Planned hubs with $X_k = 1$ and $r_{ks} \leq \chi$ are activated with pre-positioned inventory $q_k = R_k \cdot \kappa_k$; if none qualify, the safest hub is forced active. The anchor-proximity order $\pi_k = (k^{(1)}, k^{(2)}, \dots)$, ranking all hubs by ascending distance from hub k , is pre-computed once (chromosome-independent).

Table 2. Heuristic weight semantics.

Index	Name	Used in	High value means
w_0	Demand urgency	Demand sorting	Prioritise high $\lambda \cdot D$ demands first
w_1	Hub speed	Hub scoring	Prefer faster-to-reach hubs
w_2	Hub capacity	Hub scoring	Prefer hubs with more residual inventory
w_3	Demand isolation	Demand sorting	Prioritise demands with fewer reachable hubs
w_4	Hub type bonus	Hub scoring	Prefer planned over reactive hubs
w_5	Pass-1 window depth	Hub selection	Larger fraction of anchor-proximate hubs tried; favours planned over reactive

Step 2 – Demand Prioritisation. Each demand node i receives a score

$$\text{Score}_i = w_0 \widetilde{\text{urgency}}_i + w_3 \widetilde{\text{isolation}}_i - w_1 \widetilde{\text{dist}}_i + \varepsilon_i, \quad (17)$$

where $\text{urgency} = \lambda_{is} D_{is}$, $\text{isolation} = 1/|\mathcal{H}_{\text{act},i}|$, dist = minimum travel time to any active hub, each normalised to $[0, 1]$, and $\varepsilon_i \sim \mathcal{N}(0, \sigma)$ prevents premature convergence. Demands are processed in descending score order.

Step 3 – Tiered Hub Assignment. For each demand i , the trial order is π_{A_i} — all hubs sorted by proximity to the anchor hub A_i . Let $K = \max(1, \lceil w_5 \cdot |\mathcal{H}| \rceil)$.

- *Pass 1:* Among the K closest-to-anchor hubs that are active, reachable, and have positive residual inventory, select the one maximising $\text{HubScore}(k) = w_1/\tau_{ikm^*} + w_2 \cdot \text{residual}_k + w_4 \cdot \mathbf{1}[X_k = 1]$.
- *Pass 2 (overflow):* If Pass 1 fails, select the first active, reachable hub from the remaining $|\mathcal{H}| - K$ candidates (capacity violation permitted).
- *Reactive fallback:* If both passes fail, open the nearest safe inactive hub at cost F_{ks}^a ; irrecoverable infeasibility accumulates as constraint violation.

Step 4 – Supply Routing and Objective Accumulation. Origins are greedily assigned to the most under-supplied active hub. Surplus inventory is redistributed via inter-hub trans-shipment with discount α . Finally, Z_1 and Z_2 are accumulated as probability-weighted expectations over all $s \in \mathcal{S}$.

Algorithm ?? formalises the decoder for a single scenario.

3.4 Genetic Operators and Selection

The mixed-type chromosome requires mode-specific operators applied per segment. *Selection* uses binary tournament on Pareto rank with crowding distance as a tiebreaker. *Crossover* ($p_c = 0.9$) applies Uniform Crossover to $\mathbf{X}$ and $\mathbf{A}$, and Simulated Binary Crossover (SBX, $\eta_c = 20$) to $\mathbf{R}$ and $\mathbf{W}$. *Mutation* ($p_m = 1/|\mathbf{C}|$, where $|\mathbf{C}| = 2|\mathcal{H}| + |\mathcal{I}| + 6$) applies Bit-flip to $\mathbf{X}$, random-reset to $\mathbf{A}$, and Polynomial mutation ($\eta_m = 20$) to $\mathbf{R}$ and $\mathbf{W}$.

Constraint handling follows the constrained-dominance principle [6]: feasible solutions ($\text{CV} = 0$) always dominate infeasible ones; among infeasible solutions, lower CV dominates.

Algorithm 2 Priority-Based Decoder (single scenario s)

Require: Chromosome $\mathbf{C} = (\mathbf{X}, \mathbf{R}, \mathbf{A}, \mathbf{W})$, scenario s , instance data

Ensure: (Z_1^s, Z_2^s, CV^s)

```
1: Activate  $\mathcal{H}_{\text{act}} \leftarrow \{k : X_k = 1, r_{ks} \leq \chi\}$ ; set  $q_k \leftarrow R_k \kappa_k$ 
2: if  $\mathcal{H}_{\text{act}} = \emptyset$  then
3:   Force-activate  $\arg \min_k r_{ks}$  among  $\{k : X_k = 1\}$ 
4: end if
5: Score demand  $i$  by Eq. (17); sort demands in descending score order
6: for each demand  $i$  (in priority order) do
7:    $K \leftarrow \max(1, \lceil w_5 |\mathcal{H}| \rceil)$ ; let trial order  $= \pi_{A_i}$  (hubs by proximity to anchor  $A_i$ )
8:   Pass 1:  $k^* \leftarrow \arg \max \text{HubScore}(k)$  among first  $K$  active, reachable hubs with residual  $> 0$ 
9:   if Pass 1 succeeds then
10:    Assign  $i \rightarrow k^*$ ; deduct from residual inventory of  $k^*$ 
11:   else if any active reachable hub remains then
12:    Pass 2: assign  $i$  to first such hub (capacity violation allowed)
13:   else
14:    Reactive fallback: open nearest safe inactive hub; add cost  $F_{ks}^a$ ;  $CV^s += 1$ 
15:   end if
16: end for
17: Assign origins greedily; redistribute surplus via inter-hub trans-shipment (discount  $\alpha$ )
18: Accumulate  $Z_1^s, Z_2^s$ 
19: return  $(Z_1^s, Z_2^s, CV^s) = 0$ 
```

Elitist selection with Hamming diversity. At each generation, the combined parent–offspring pool of $2N$ individuals is trimmed to N via fast non-dominated sorting. When a partial Pareto front must be split, tiebreaking proceeds as: (1) Pareto rank, (2) crowding distance, (3) minimum Hamming distance to the nearest neighbour in $\mathbf{X}$ -space. This third criterion preserves genotypic diversity in hub configuration space and counters premature convergence. Hamming distances are recomputed each generation over the $2N$ pool in $O(N^2|\mathcal{H}|)$ time.

4 Computational Experiments

4.1 Experimental Setup

Implementation. PB-NSGA is implemented in C++17. Data generation and result analysis are performed in Python 3. All experiments run on a standard workstation (Intel Core i7, 16 GB RAM).

Algorithm parameters. Population $N = 100$; generations $G = 200$; $p_c = 0.9$; $\eta_c = \eta_m = 20$; $p_m = 1/|\mathbf{C}|$ where $|\mathbf{C}| = 2|\mathcal{H}| + |\mathcal{I}| + 6$; decoder noise $\sigma = 0.05$; *Pass-1* window $K = \max(1, \lceil w_5 \cdot |\mathcal{H}| \rceil)$ (evolved per individual). The safety threshold is $\chi = 0.7$; discount $\alpha = 0.6$; $\gamma = 3.0$ kg/person; $\lambda_0 = 0.8$.

Performance metrics. Each algorithm is executed across 20 independent runs per instance; all indicators are reported as mean $\pm$ standard deviation, with statistical significance assessed via the Wilcoxon rank-sum test ($\alpha = 0.05$). We employ three complementary Pareto-front quality indicators:

- **Hypervolume (HV)** [?]: volume of objective space dominated by the approximation set, bounded by reference point $\mathbf{r} = (1.1, 1.1)$ in the $[0, 1]^2$ normalised space (nadir of the combined front). A larger HV indicates better convergence and spread.
- **IGD⁺** [?]: average modified Hausdorff distance from each reference point to its closest solution in the approximation set. For small instances (AP10, AP20, AP25, AP40, CV-Small), the reference set is the *exact true Pareto front* obtained by complete enumeration (Section 4.3); for larger instances, it is the combined non-dominated front pooled across all 20 runs and all algorithms. A smaller IGD⁺ indicates closer proximity to the true front.
- **Set Coverage (C-metric)** [?]: $C(A, B)$ denotes the fraction of solutions in B dominated by at least one solution in A . We report $C(\text{PB-NSGA}, \text{NSGA-II})$ and $C(\text{NSGA-II}, \text{PB-NSGA})$; a high $C(A, B)$ with low $C(B, A)$ indicates algorithm A dominates a larger portion of B ’s Pareto front.

4.2 Dataset Description

Benchmark Instances (HLP $\rightarrow$ DRND Transformation) We adapt two widely-used hub location benchmarks into DRND format. The **AP** set [7] provides Euclidean instances of sizes $n \in \{10, 20, 25, 40, 50, 100\}$; the **TR81** dataset [11] is a real inter-city distance and flow matrix for 81 Turkish provincial capitals.

Each benchmark instance is transformed via the following procedure:

1. **Role assignment.** Nodes are ranked by a safety-flow score $s_i = f_i(1 - \bar{r} - 0.3\tilde{f}_i)$, where f_i is total incident flow and $\tilde{f}_i$ its normalised value. Top- p nodes ($p = \lfloor 0.3n \rfloor$) are designated hubs; the remaining high-outflow nodes become origins/suppliers.
2. **Scenario generation.** Three flood scenarios (mild, severe, extreme) are assigned probabilities $(\pi_1, \pi_2, \pi_3) = (0.60, 0.30, 0.10)$. For each scenario, a random epicenter is drawn; road disruption penalties decay exponentially with distance ($\sigma \in [0.25, 0.35] \times d_{\max}$).
3. **Calibration.** Distances are normalised to a 500 km scale. Supply is set to 1.5–2.5 $\times$ total demand. Hub fixed costs are proportional to average road distance (km-units). Three transport modes operate concurrently (Table 3).

Central Vietnam Case Study Instances Two instances are constructed to model flood relief operations across the **Vu Gia – Thu Bn – Perfume** river basin, spanning four disaster-prone provinces: à Nng, Qung Nam, Tha Thiên-Hu, and Qung Ngãi.

- **CV-Small:** $|\mathcal{I}| = 20$, $|\mathcal{H}| = 5$, $|\mathcal{J}| = 2$, $|\mathcal{S}| = 3$ – district-level planning instance.

Table 3. Transport mode parameters used for all instances.

Mode	Vehicle	Speed (km/h)	Cost (\$/km)	Capacity (pers.)	Disruption
0	Truck	35	2.0	60	Road-dependent
1	Motorboat	25	5.0	25	Low (flood-resilient)
2	Helicopter	150	40.0	10	None (always available)

- **CV-Large:** $|\mathcal{I}| = 100$, $|\mathcal{H}| = 20$, $|\mathcal{J}| = 12$, $|\mathcal{S}| = 3$ – commune-level regional instance.

Node georeferencing. Demand nodes are georeferenced to real administrative units (districts or communes) from four provinces. Base population follows a coastal-proximity model, $P_i \approx 500 + 6,500 \cdot f_{\text{coast}}$, reflecting the densely-settled coastal plains of Central Vietnam. Candidate hub locations are selected from logistics facilities, rescue stations, and military depots with confirmed multi-modal access. Supply origins are drawn from seaports (à Nng, Dung Quat, Chu Lai, Sa Kỳ), river ports (Thun An), and highland border logistics nodes.

Multi-criteria auxiliary risk. Each node u receives a static intrinsic flood-risk score $r_u^a \in [0, 1]$, computed as the weighted sum of four geographic criteria following [?]:

$$r_u^a = 0.25 C_1(u) + 0.35 C_2(u) + 0.20 C_3(u) + 0.20 C_4(u), \quad (18)$$

where C_1 is a topographic inundation index (decreasing with elevation, proxied by longitude relative to the highland–coastal gradient); C_2 is hydrological proximity to five river systems (Vu Gia, Thu Bn, Perfume, Trng Giang, Trà Bng) modelled via Gaussian decay ($\sigma = 18$ km); C_3 is coastal storm-surge exposure ($\sigma = 40$ km); and C_4 is river-delta accumulation risk centred on five historically inundated deltaic zones.

Risk intervals and scenario generation. Rather than assigning a single risk value per node, each node is assigned a risk interval $[r_u^{\min}, r_u^{\max}]$ whose width and ceiling scale monotonically with r_u^a :

$$r_u^{\min} = 0.05 + 0.20 r_u^a, \quad r_u^{\max} = r_u^a + 0.15(1 - r_u^a).$$

A safe inland node ($r_u^a = 0.10$) thus spans $[0.07, 0.22]$, while a highly exposed delta node ($r_u^a = 0.80$) spans $[0.21, 0.83]$. Three scenarios (mild, severe, extreme) are generated with probabilities (0.60, 0.30, 0.10). Each scenario samples $n_s \in \{1, 2, 3\}$ flood epicenters from demand nodes with probability proportional to r_u^a ; scenario risk for node u is then drawn from its interval, modulated by Gaussian exposure decay ($\sigma_e = 85$ km) from the epicenters. Road link (u, v) is disrupted with probability $\min(0.97, \beta_s \cdot \bar{r}_{uv})$, where $\beta_s \in \{0.25, 0.55, 0.88\}$ for mild, severe, and extreme events; water and air modes remain available under all scenarios.

Transport and capacity calibration. Road travel times combine Haversine distance with a fixed tortuosity factor (1.35, reflecting Central Vietnam’s mountain roads) and a terrain multiplier ranging from 1.0 at the coast to 1.8

in the deep highlands. Demand volumes are scenario-specific: $D_{is} = P_i \cdot (0.05 + 0.85 r_{is}) \cdot \mu_s$, where severity multipliers $\mu_s \in \{1.0, 1.8, 2.8\}$ capture the escalating relief burden. Last-mile distribution cost is modelled by the Daganzo continuum-approximation formula [?] with scenario-specific circuitry $\phi \in \{0.57, 0.70, 0.85\}$ for the three scenarios. Hub capacities are calibrated post-generation so that aggregate installed capacity is $3\text{--}6\times$ the worst-case total demand, with each hub’s share scaled by its terrain factor to account for the larger staging areas feasible in flatter coastal zones.

4.3 Experiment 1: Algorithm Efficiency on HLP Benchmarks

Comparison Algorithms We compare **PB-NSGA** against the standard **NSGA-II** [6] as a direct baseline. NSGA-II uses the same population size ($N = 100$), generations ($G = 200$), and genetic operators but replaces the priority-based decoder with a direct full variable encoding (all second-stage routing variables). This comparison isolates the contribution of the priority-based decoding scheme. Both algorithms are seeded identically across 20 runs.

Ground-truth fronts via complete enumeration. To provide an objective quality reference for small benchmark instances, we apply a *complete enumeration* procedure that exhausts all $2^{|\mathcal{H}|}$ hub opening configurations. For each configuration $\mathbf{X}$, the second-stage sub-problem is solved via a structured heuristic (multiple candidate $\mathbf{R}$, $\mathbf{A}$, $\mathbf{W}$ settings) to approximate the true objective values, and only non-dominated solutions are retained. A 5-minute time limit is imposed per instance. This yields an exact (or near-exact) true Pareto front for AP10, AP20, AP25, AP40, and CV-Small ($|\mathcal{H}| \leq 10$), which serves as the reference set for IGD^+ computation on these instances.

Results Table 4 reports HV, IGD^+ , Pareto front size, and CPU time for the AP and TR81 benchmark instances. Values are mean $\pm$ standard deviation over 20 runs.

Discussion. For small instances (AP10–AP40, CV-Small), IGD^+ is measured against the exact true Pareto front from complete enumeration, providing an absolute quality reference independent of the competing algorithm. For AP50, AP100, and TR81 – where exhaustive enumeration is infeasible – the combined non-dominated front across all 20 runs and both algorithms serves as the reference. We also report C-metric pairwise between PB-NSGA and NSGA-II; $C(A, B) \gg C(B, A)$ indicates that A ’s front dominates a substantially larger fraction of B ’s solutions. The anchor-hub mechanism ($\mathbf{A}$ chromosome) is expected to be particularly beneficial on larger instances, where the full variable encoding of standard NSGA-II disperses search effort across an exponentially larger space.

4.4 Sensitivity Analysis

To assess how key operational parameters affect solution quality, we conduct one-at-a-time sensitivity experiments on the CV-Small instance using PB-NSGA

Table 4. Experiment 1: Algorithm efficiency on HLP benchmark instances (mean $\pm$ std, 20 runs). Best mean per instance in **bold**. C-metric: $C(\text{PB-NSGA}, \text{NSGA-II}) / C(\text{NSGA-II}, \text{PB-NSGA})$.

Instance	Algorithm	HV (norm.)	IGD ⁺	#Pareto	Time (s)	C-metric
AP10	PB-NSGA	—	—	—	—	— / —
	NSGA-II	—	—	—	—	— / —
AP20	PB-NSGA	—	—	—	—	— / —
	NSGA-II	—	—	—	—	— / —
AP25	PB-NSGA	—	—	—	—	— / —
	NSGA-II	—	—	—	—	— / —
AP40	PB-NSGA	—	—	—	—	— / —
	NSGA-II	—	—	—	—	— / —
AP50	PB-NSGA	—	—	—	—	— / —
	NSGA-II	—	—	—	—	— / —
AP100	PB-NSGA	—	—	—	—	— / —
	NSGA-II	—	—	—	—	— / —
TR81	PB-NSGA	—	—	—	—	— / —
	NSGA-II	—	—	—	—	— / —

with 10 independent runs per configuration, varying one parameter at a time while holding others at their baseline values ($\chi = 0.7$, $\pi_3 = 0.10$, transport costs at baseline).

Safety threshold χ . We vary $\chi \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$. A stricter (lower) threshold restricts the set of safe hubs under disruption, forcing more reactive openings and increasing Z_1 ; a lenient (higher) threshold allows more hubs to remain operational at higher risk, elevating Z_2 .

Scenario probability weight π_3 . We test $\pi_3 \in \{0.05, 0.10, 0.15, 0.20\}$ (with π_1 and π_2 adjusted proportionally), quantifying how the optimal pre-positioning strategy shifts as planners assign greater probability to worst-case extreme-flood events.

Transportation cost multiplier β . All per-km costs are scaled by $\beta \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$, testing the modal shift from road to water/air transport as surface-route costs increase.

4.5 Experiment 2: Case Study – Central Vietnam Flood Relief

Comparison Algorithms For the Vietnam case study we include two additional multi-objective solvers alongside PB-NSGA:

- **NSGA-II** [6]: standard baseline (same encoding as Experiment 1).
- **NSMA** [?]: Non-dominated Sorting Memetic Algorithm, a state-of-the-art hybrid incorporating local search phases.

All three algorithms use identical budget ($N = 100$, $G = 200$) and are evaluated over 20 seeds per instance.

Table 5. Sensitivity of mean HV to key operational parameters (CV-Small, 10 runs each; baseline values in parentheses).

Parameter	Level	HV (mean)
Safety threshold χ	0.5	–
	0.6	–
	0.7 (<i>baseline</i>)	–
	0.8	–
	0.9	–
Extreme probability π_3	0.05	–
	0.10 (<i>baseline</i>)	–
	0.15	–
	0.20	–
Cost multiplier β	0.5	–
	0.75	–
	1.0 (<i>baseline</i>)	–
	1.5	–
	2.0	–

Results Figure 1 shows the combined Pareto fronts (pooled across 20 seeds) for CV-Small and CV-Large. Table 5 summarises HV, IGD⁺, and Pareto front size.

Table 6. Experiment 2: Case study results – Central Vietnam instances (mean $\pm$ std, 20 runs). Best mean per instance in **bold**. C-metric column: $C(\text{PB-NSGA}, \text{Competitor}) / C(\text{Competitor}, \text{PB-NSGA})$.

Instance	Algorithm	HV (norm.)	IGD ⁺	#Pareto	C-metric
CV-Small	PB-NSGA	–	–	–	–
	NSGA-II	–	–	–	– / –
	NSMA	–	–	–	– / –
CV-Large	PB-NSGA	–	–	–	–
	NSGA-II	–	–	–	– / –
	NSMA	–	–	–	– / –

Managerial Insights The Pareto front of CV-Small reveals a convex cost-deprivation trade-off. The shape reflects the non-linear structure of the deprivation objective: reducing Z_2 in the high-cost region of the frontier requires disproportionate investment in additional hub capacity and air-/water-mode transport. Humanitarian agencies with limited budgets should identify their operating point on this frontier rather than minimising cost alone.

Fig. 1. Combined Pareto fronts for CV-Small (left) and CV-Large (right). Each front pools all non-dominated solutions across 20 seeds per algorithm.

Network topology analysis highlights scenario-driven structural shifts. Under the *mild* scenario, road transport from a moderate set of candidate hubs provides adequate coverage. Under the *extreme* scenario, a subset of hubs exceeds the safety threshold ($r_{ks} > \chi$) and becomes unavailable, triggering automatic modal substitution toward waterway and helicopter routes. The remaining active hubs absorb greater pre-positioned inventory, partially compensated by inter-hub trans-shipment. This dynamic adaptation confirms the operational value of the two-stage stochastic formulation: a deterministic model calibrated to a single representative scenario would systematically under-invest in resilient transport modes and backup hub capacity.

For CV-Large, the anchor-hub mechanism encoded in **A** proves particularly valuable. Commune-level demand nodes form distinct geographic clusters aligned with river basins and coastal plains; the **A** chromosome implicitly partitions the search space into hub catchment areas, enabling PB-NSGA to maintain diversity across 20 hub candidates while allocating 100 demand nodes efficiently.

5 Conclusion and Future Work

This paper introduced PB-NSGA, a Priority-Based Nondominated Sorting Genetic Algorithm, to solve the Multi-Objective Incomplete Hub Location Network Design Problem (MO-IHLNDP) for disaster relief logistics. The model jointly minimises expected logistics cost (Z_1) and expected maximum deprivation cost (Z_2) under stochastic flood disruptions, capturing multi-modal transport, hub safety constraints, reactive hub opening, and inter-hub trans-shipment with economies of scale.

Three methodological contributions underpin this work. First, the chromosome representation encodes only first-stage strategic decisions (**X**, **R**, **A**, **W**), reducing the search space from $O(|\mathcal{S}| \times |\mathcal{V}|)$ to $O(|\mathcal{H}|)$ while the heuristic decoder reconstructs all second-stage assignments. Second, the anchor-hub mechanism (**A** chromosome) biases demand assignment toward geographically coherent hub neighbourhoods, which proves especially powerful for large instances and spatially clustered demand. Third, Hamming diversity in elitist selection preserves genotypic variety in hub configuration space, countering premature convergence to narrow Pareto fronts. To benchmark solution quality objectively, ground-truth Pareto fronts for small instances are obtained by complete enumeration of all hub configurations, providing exact IGD⁺ reference sets.

The Central Vietnam case study, constructed from real geographic and demographic data across four flood-prone provinces, illustrates the practical value of the framework. The two-stage stochastic model responds to infrastructure disruption by dynamically shifting to water and air transport and redistribut-

ing inventory, demonstrating that deterministic planning would systematically under-invest in network resilience.

Limitations. The model assumes a fixed discrete scenario set; continuous risk distributions are not yet supported. The decoding heuristic is greedy and may miss globally optimal demand assignments for highly constrained instances.

Future work will extend the approach to rolling-horizon multi-period relief operations and incorporate real-time sensor feeds for online re-optimization. The complete enumeration baseline naturally motivates exact solver development (branch-and-cut or Benders decomposition) for medium-scale instances to tighten IGD⁺ reference fronts and strengthen convergence guarantees.

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